

Mendelian Susceptibility to Mycobacterial Disease due to Complete ISG15 Deficiency

Comprehensive Disease Characterization Report

Disease: Complete ISG15 Deficiency (Immunodeficiency 38 with basal ganglia calcification)
MONDO: MONDO:0014502 | **OMIM (disease):** #616126 (IMD38) | **OMIM (gene):** 147571
Orphanet: ORPHA:319563 | *Category:** Mendelian, autosomal recessive inborn error of immunity

Summary (Answer to the Research Question)

Complete ISG15 deficiency is a rare **autosomal recessive inborn error of immunity** caused by biallelic loss-of-function variants in **ISG15** (interferon-stimulated gene 15; a ubiquitin-like modifier). It is a **mixed syndrome** that uniquely combines two mechanistically opposite immune defects: (1) **Mendelian Susceptibility to Mycobacterial Disease (MSMD)** — recurrent infection by low-virulence mycobacteria (BCG vaccine strain, environmental mycobacteria) caused by loss of **secreted ISG15**, which is a non-redundant inducer of **IFN- γ** from NK and T lymphocytes; and (2) a **type I interferonopathy** — because **intracellular ISG15 is required to stabilize USP18**, the principal negative regulator of type I IFN (IFN- α/β) signaling. Loss of intracellular ISG15 $\rightarrow$ USP18 degradation $\rightarrow$ uncontrolled amplification of IFN- α/β responses $\rightarrow$ **basal ganglia/intracranial calcification, autoinflammation, and necrotizing/fibrotic skin lesions**. Notably, human ISGylation is largely **redundant for antiviral defense** (patients are not unusually susceptible to viruses), which distinguishes human from mouse biology. Management splits along the two mechanisms: **anti-mycobacterial therapy** for infection and **JAK1/2 inhibition** (baricitinib, ruxolitinib) for the interferonopathy.

Key	primary	references:	Bogunovic	et	al.,	<i>Science</i>	2012
(							);
	P	22859821					
Zhang		et		al.,		<i>Nature</i>	2015
(							);
	P	25307056					
Martin-Fernandez			et		al.		2020
(							);
	P	32402279					
Sazeides		et			al.		2026
(							).
	P	41743834					

1. Disease Information

- **Overview:** An autosomal recessive primary immunodeficiency / inborn error of immunity in which complete absence of the ISG15 protein produces a combined phenotype of susceptibility to weakly virulent mycobacteria and a monogenic type I interferonopathy with intracranial calcification and inflammatory skin disease.
- **Key identifiers:**
- **MONDO:** MONDO:0014502
- **OMIM disease:** #616126 "Immunodeficiency 38 with basal ganglia calcification; IMD38"
- **OMIM gene (ISG15):** *147571
- **Orphanet:** ORPHA:319563 ("Immunodeficiency due to ISG15 deficiency")
- **HGNC:** HGNC:4053 (ISG15) | **NCBI Gene:** 9636 | **Ensembl:** ENSG00000187608 | **UniProt:** P05161
- **ICD-10:** D84.8 (other specified immunodeficiencies); **ICD-11:** 4A00.x (primary immunodeficiencies); basal ganglia calcification coded separately.
- **MeSH:** related terms — "Immunologic Deficiency Syndromes"; "Interferon-Stimulated Gene 15"; "Ubiquitins."
- **Synonyms / alternative names:** Complete ISG15 deficiency; ISG15 deficiency; Immunodeficiency 38 (IMD38); Immunodeficiency 38 with basal ganglia calcification; Mendelian susceptibility to mycobacterial disease due to ISG15 deficiency; ISG15-related type I interferonopathy.

- **Information source:** Aggregated from **individual patient case reports and small kindreds** (the disease is ultra-rare; total reported patients are in the low dozens worldwide), synthesized with **disease-level resources** (OMIM, Orphanet). Not derived from EHR/population cohorts.

2. Etiology

- **Primary cause (genetic):** Biallelic (homozygous or compound heterozygous) **loss-of-function variants in ISG15** producing complete absence of ISG15 protein and ISGylation. This is a monogenic Mendelian disorder.
- **Genetic risk factors:** The causal variants themselves; **consanguinity** is a major contributor (most families are consanguineous, from Turkey, Iran, Middle East, China, Brazil). No common susceptibility loci or modifier genes are established; MPO variants co-segregating in one kindred were shown to be non-pathogenic
([P 37984483](#)).
- **Environmental risk factors / triggers:** **BCG vaccination** (live attenuated *Mycobacterium bovis*) is the principal environmental trigger of the infectious phenotype — most patients present after BCG with regional or disseminated BCG disease (BCGitis/BCGosis). **Environmental (non-tuberculous) mycobacteria** are the other trigger. There is no strong sex or occupational exposure signal.
- **Protective factors:** None established genetically. From a preventive standpoint, **avoidance of BCG vaccination** in known-affected families prevents BCG disease.
- **Gene–environment interaction:** The genetic lesion (low IFN- γ inducibility) is **clinically unmasked by mycobacterial exposure** (BCG/environmental mycobacteria). Individuals never exposed to mycobacteria may present instead with the neurologic or dermatologic (interferonopathy) phenotype, which is exposure-independent.

3. Phenotypes

The syndrome has **three co-dominant clinical presentations** (infectious, neurologic, dermatologic); individual patients may show one, two, or all three.

Phenotype	Type	HPO suggestion	Onset	Frequency/ notes
Susceptibility to BCG / environmental mycobacteria (BCGitis, BCGosis, lymphadenitis)	clinical/ infectious	HP:0002754 (osteomyelitis), HP:0032262 (atypical mycobacterial infection), HP:0004432 (abnormal mycobacterial immunity)	Infancy/ childhood (post-BCG)	Common in exposed patients
Intracranial/basal ganglia calcification	imaging/ neurologic	HP:0002514 (cerebral calcification), HP:0007146 (bilateral basal ganglia calcification)	Childhood; often asymptomatic detection	Frequent; may be subclinical
Seizures	clinical sign	HP:0001250 (seizure)	Variable	Reported subset
Necrotizing / ulcerative skin lesions	physical manifestation	HP:0200041 (skin ulcer), HP:0000951 (abnormal skin morphology)	Childhood– adult	Third phenotype (P 32402279)
Skin fibrosis / impaired wound healing	physical manifestation	HP:0100699 (scarring), HP:0001075 (atrophic scars)	Chronic	P 41743834
Recurrent pneumonia / inflammatory lung disease	clinical	HP:0006532 (recurrent pneumonia)	Variable	P 37984483
Elevated blood type I IFN signature (ISG upregulation)	laboratory abnormality	HP:0031381 (increased circulating interferon)	Constitutive	Universal (biomarker)

Phenotype	Type	HPO suggestion	Onset	Frequency/ notes
Low mycobacterium-induced IFN- γ production	laboratory abnormality	HP:0032218 (decreased IFN- γ production)	Constitutive	Universal in vitro (P 22859821)

- **Severity:** Variable; ranges from asymptomatic calcification to life-threatening disseminated mycobacterial disease. The interferonopathy is generally **milder than USP18 deficiency** (which is lethal in infancy).
- **Progression:** Intracranial calcification tends to be **stable/slowly progressive**; skin disease can be chronic-relapsing; infections are **episodic** and exposure-linked.
- **Quality-of-life impact:** Chronic skin ulcers, recurrent infections, and (in some) neurologic sequelae/seizures impair daily functioning; formal QoL instrument data are not available for this ultra-rare disease.

4. Genetic / Molecular Information

- **Causal gene:** ISG15 (ISG15 ubiquitin-like modifier), chromosome **1p36.33**; encodes a ~17 kDa precursor cleaved to a **15 kDa** mature ubiquitin-like protein bearing a C-terminal LRLRGG motif used for conjugation (ISGylation).
- **Pathogenic variants:**
- **Type/class:** Predominantly **null / loss-of-function** — nonsense, frameshift, and destabilizing missense variants, all resulting in **complete deficiency**. Reported examples include **p.Leu28Gln (c.83T>A)** — a destabilizing missense ($\Delta\Delta G -2.4$ kcal/mol;

([P 39365299](#));

p.Tyr140* (Y140X) nonsense

([P 37984483](#));

and various frameshift/nonsense alleles in the original Turkish/Iranian kindreds

([P 22859821](#))

([P 25307056](#)).

- **Classification:** Pathogenic/likely pathogenic per ACMG/AMP (loss-of-function is the established disease mechanism; PVS1-supporting).
- **Allele frequency:** Individually **ultra-rare/private**; homozygosity typically arises through consanguinity. Not present at appreciable frequency in gnomAD.
- **Origin: Germline**, biallelic. No somatic mechanism.
- **Functional consequence: Loss of function** — absence of both free ISG15 (secreted) and conjugated ISG15 (ISGylation), and secondary loss of USP18 stabilization.
- **Modifier genes:** None validated. Digenic contributions have been excluded in reported kindreds (e.g., MPO co-variants non-pathogenic,

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- **Epigenetics:** No disease-specific methylation/chromatin defect described; however, the disease produces a strongly altered **transcriptional (IFN-driven) program** rather than a primary epigenetic lesion.
- **Chromosomal abnormalities:** None; this is a single-gene disorder, not a copy-number/structural syndrome.

5. Environmental Information

- **Environmental factors / infectious agents (central to the infectious phenotype):**
- **Mycobacterium bovis BCG** (vaccine strain) — NCBI Taxon 33892 — principal trigger.
- **Environmental / non-tuberculous mycobacteria** (e.g., *M. avium* complex) — low-virulence mycobacteria.
- Predisposition is essentially restricted to **weakly virulent mycobacteria**; classical *M. tuberculosis* and *Salmonella* susceptibility is variable, as in other MSMD.
- **Lifestyle factors:** Not applicable/none identified.
- **Non-infectious environmental toxins/radiation:** No established role.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

Branch A — Infectious/MSMD phenotype: 1. Biallelic ISG15 loss-of-function variant **leads to** absence of ISG15 protein. 2. Absence of **secreted (extracellular) ISG15** from leukocytes (especially granulocytes/neutrophils) and epithelial cells **results in** loss of a non-redundant IFN- γ -inducing signal. 3. Loss of ISG15 **binding to the LFA-1 integrin (CD11a/CD18; α L β 2) receptor** on NK and T cells **results in** failure to trigger IFN- γ release (Swaim et al.,

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4. Reduced mycobacterium-induced **IFN- γ production leads to** impaired macrophage activation and killing of intracellular mycobacteria. 5. This **results in** susceptibility to BCG and environmental mycobacteria (MSMD). (demonstrated:

P 22859821

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Branch B — Neurologic/dermatologic interferonopathy: 1. Absence of **intracellular ISG15 leads to** failure to stabilize **USP18** (ISG15 normally protects USP18 from proteasomal degradation). 2. Loss of USP18 **results in** failure to sterically block JAK1 at IFNAR2 → **sustained/amplified IFN- α/β (JAK-STAT/ISGF3) signaling**. 3. Chronic type I IFN over-activity **leads to** an autoinflammatory state resembling Aicardi-Goutières syndrome — **basal ganglia/intracranial calcification** (inferred to arise from IFN-driven vascular/glial injury), autoinflammation. 4. In skin, IFN-I signaling in **keratinocytes, endothelia, and dermal monocytes/macrophages results in** apoptosis, altered macrophage polarization, epithelial-to-mesenchymal transition, and myofibroblast activation → **necrotizing and fibrotic skin lesions and impaired wound healing**. (demonstrated:

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Category checklist

- **Molecular pathways:** Type I IFN **JAK–STAT (JAK1/TYK2 → STAT1/STAT2/IRF9/ISGF3)** signaling (overactive); IL-12/IFN- γ axis (underactive); **secreted ISG15 → LFA-1 integrin (ITGAL/ITGB2, CD11a/CD18) → IFN- γ release** (GO:0005178 integrin binding); **ISG15/ISGylation** ubiquitin-like conjugation cascade (E1 UBE1L/UBA7, E2 UBCH8/UBE2L6, E3 HERC5); deISGylation by USP18.
- **Reactome/KEGG:** IFN- α/β signaling; ISG15 antiviral mechanism.
- **Cellular processes:** Chronic **inflammation**, IFN-I-induced **apoptosis**, **EMT**, myofibroblast activation, altered macrophage polarization.
- **Protein dysfunction:** Complete **loss of function** of ISG15; secondary **destabilization/loss of USP18**. ISG15 is a **diubiquitin-like protein** comprising two tandem ubiquitin-like (Ubl) domains with a C-terminal LRLGG conjugation motif (InterPro IPR000626 ubiquitin-like domain; Pfam PF00240; UniProt P05161). Structural studies of the USP18–ISG15 complex show USP18 recognizes only the **C-terminal Ubl domain** of ISG15 via a small hydrophobic interface, explaining USP18's ISG15-specific (not ubiquitin) deconjugation activity and the mutual stabilization relationship (mouse USP18–ISG15 crystal structures,

(P 28165509

PDB e.g. 5CHV/5CHW). Destabilizing missense variants (e.g., p.Leu28Gln) increase protein flexibility/ $\Delta\Delta G$ and abolish function

(P 39365299).

- **Immune system involvement:** Combined defect — **immunodeficiency** (hypomorphic IFN- γ , antimycobacterial) plus **autoinflammation** (excess IFN- α/β).
- **Tissue damage mechanisms:** IFN-I-mediated apoptosis, fibrosis, dystrophic calcification.
- **Molecular profiling:** Patient RNA-seq shows strong **upregulation of interferon-stimulated genes** and JAK/STAT hyperactivation

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spatial transcriptomics localizes IFN-I signatures to epidermis/dermal myeloid cells

(P 41743834).

- **Suggested GO terms:** GO:0060337 (type I interferon signaling pathway), GO:0032479 (regulation of type I interferon production), GO:0032020 (ISG15-protein conjugation), GO:0006954 (inflammatory response), GO:0060333 (interferon-gamma-mediated signaling).
 - **Suggested CL terms:** CL:0000775 (neutrophil), CL:0000623 (NK cell), CL:0000576 (monocyte), CL:0000235 (macrophage), CL:0000312 (keratinocyte), CL:0000057 (fibroblast/myofibroblast).
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7. Anatomical Structures Affected

- **Organ level:**
 - **Brain** — basal ganglia and other intracranial sites (calcification). UBERON:0002420 (basal ganglion); UBERON:0000955 (brain).
 - **Skin** — necrotizing/fibrotic lesions, ulcers. UBERON:0002097 (skin of body); UBERON:0001003 (epidermis).
 - **Lymph nodes / lymphatic system** — mycobacterial lymphadenitis. UBERON:0000029 (lymph node).
 - **Lungs** — recurrent pneumonia/inflammatory lung disease. UBERON:0002048 (lung).
 - **Body systems:** Immune, nervous (CNS), integumentary; secondarily lymphatic and respiratory.
 - **Tissue/cell level:** Epidermal **keratinocytes**, dermal **endothelium**, **monocytes/macrophages**, **neutrophils** (secretion source), **NK/T lymphocytes** (IFN- γ effectors), **fibroblasts/myofibroblasts** (fibrosis).
 - **Subcellular level (GO Cellular Component):** Cytoplasm/cytosol (GO:0005829) — site of ISGylation and JAK-STAT signaling; ISG15 conjugated to targets throughout cytoplasm and nucleus.
 - **Localization / lateralization:** Intracranial calcification is characteristically **bilateral/symmetric (basal ganglia)**; skin lesions are variable in distribution.
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8. Temporal Development

- **Onset:** Typically **infancy to childhood**. Infectious presentation follows **BCG vaccination** (first months–years of life). Interferonopathy features (calcification) may be detected incidentally in childhood; dermatologic disease can present later, including adulthood.
- **Onset pattern:** Infections are **acute/subacute and episodic**; the interferonopathy is **chronic/insidious**.
- **Progression:** Basal ganglia calcification is generally **stable or slowly progressive**; skin disease chronic-relapsing; the overall interferonopathy is **milder and more survivable** than complete USP18 deficiency (neonatal lethal).
- **Disease course/duration:** **Chronic, lifelong**. No spontaneous cure; remission of inflammation is **treatment-induced** (JAK inhibition).
- **Critical periods:** The **peri-BCG-vaccination window** (infancy) is the key window for the infectious phenotype; earlier initiation of JAK inhibition may better limit inflammatory sequelae, though CNS damage may still progress.

9. Inheritance and Population

- **Inheritance:** **Autosomal recessive** (biallelic loss-of-function).
- **Penetrance/expressivity:** Effectively complete at the biochemical level (absent ISG15), but **variable clinical expressivity** — which of the three phenotypes manifests depends partly on environmental (mycobacterial) exposure and unknown modifiers; siblings with identical variants can show divergent presentations
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- **Consanguinity / founder effects:** Strong **consanguinity** association; most kindreds are from Turkey, Iran, the Middle East, China, and Brazil. Recurrent identical variants in unrelated families suggest possible local founder alleles (e.g., c.83T>A in Brazil,

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- **Carrier frequency:** Very low; heterozygous carriers are asymptomatic. No established population carrier screening.

- **Epidemiology:** Ultra-rare (Orphanet class; prevalence <1/1,000,000). Fewer than ~several dozen patients reported worldwide since 2012. No reliable incidence/prevalence estimates.
- **Sex ratio:** No strong sex bias reported (autosomal recessive).
- **Geographic distribution:** Reported predominantly in consanguineous populations; specific variants show regional clustering.

10. Diagnostics

- **Laboratory / functional tests:**
- **Absent ISG15 protein and ISGylation** by immunoblot of patient cells (diagnostic hallmark).
- **Reduced IFN- γ production** by PBMCs/whole blood upon **BCG or BCG+IL-12** stimulation
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- **Elevated type I IFN signature** — interferon-stimulated gene (ISG) expression score in blood; hyperactivated **JAK/STAT (phospho-STAT1)** signaling
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- **Elevated type I IFN signature** — interferon-stimulated gene (ISG) expression score in blood; hyperactivated **JAK/STAT (phospho-STAT1)** signaling
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- Absence of **USP18 accumulation** in patient cells (research assay).
- **Biomarkers:** Blood **IFN signature (ISG score)** — quantified by RT-qPCR of a panel of interferon-stimulated genes (e.g., IFI27, IFI44L, IFIT1, ISG15, RSAD2, SIGLEC1) or by **SIGLEC1/CD169 surface expression on monocytes** (standard type I interferonopathy biomarkers); low mycobacterium-induced IFN- γ ; **absent ISG15 protein on Western blot** (the specific molecular hallmark). LOINC-codable functional assay: IFN- γ release after BCG $\pm$ IL-12 stimulation.
- **Imaging: CT/MRI of brain** showing **bilateral basal ganglia (and other intracranial) calcifications** — a key diagnostic clue (RadLex; overlaps AGS/SPENCD imaging).
- **Genetic testing (definitive): Whole-exome (WES) or whole-genome sequencing**, or targeted **MSMD/inborn-errors-of-immunity gene panels** including ISG15, followed by **Sanger confirmation**

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Single-gene ISG15 sequencing is diagnostic once suspected.

- **Clinical criteria / context:** Diagnosis is suspected in a child with **BCG/environmental mycobacterial disease** and/or **unexplained basal ganglia calcification** and/or **necrotizing skin lesions**, confirmed molecularly.
- **Differential diagnosis:** Other **MSMD genes** (IFNGR1/2, IL12B, IL12RB1, STAT1, IRF8, TYK2, SPPL2A) for the infectious phenotype; other **type I interferonopathies** — **Aicardi-Goutières syndrome**, **spondyloenchondrodysplasia (SPENCD)**, **USP18 deficiency**, **STAT2 gain-of-function (R148Q)** — for the neurologic/calcification phenotype

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USP18 deficiency is far more severe (neonatal lethal).

- **Screening:** **Cascade genetic testing** of relatives in known families; consider avoiding BCG in at-risk newborns.

11. Outcome / Prognosis

- **Survival/mortality:** Generally **better than USP18 deficiency** (which is lethal in infancy). Prognosis depends on control of mycobacterial infections and the severity of the interferonopathy; disseminated mycobacterial disease can be life-threatening. No formal survival statistics exist (ultra-rare).
- **Morbidity/function:** Chronic morbidity from **recurrent infections, seizures/neurologic sequelae, and chronic skin ulceration/fibrosis**; disability is variable.
- **Complications:** Disseminated BCG disease, mycobacterial lymphadenitis/osteomyelitis, recurrent pneumonia, neurologic impairment from calcification/inflammation, and multi-organ fibrosis (skin, lung).
- **Prognostic factors:** Extent of mycobacterial dissemination; magnitude of IFN-I signature; timing of JAK-inhibitor initiation; degree of CNS involvement.

- **Recovery potential:** Infections are treatable; the interferonopathy is controllable but not curable pharmacologically; established CNS damage may be irreversible.

12. Treatment

Therapy is **split along the two mechanisms** (NCIT concepts noted):

- **Anti-mycobacterial therapy** for BCG/environmental mycobacterial disease (multidrug regimens; e.g., rifampin, isoniazid, ethambutol, macrolides per species). NCIT: antimycobacterial agents.
- **Adjunctive IFN- γ :** Recombinant **IFN- γ (IFN- γ 1b)** may be considered for refractory mycobacterial infection, as in other MSMD with hypomorphic IFN- γ . NCIT:C1471 (Interferon Gamma).
- **JAK1/2 inhibitors** for the type I interferonopathy (skin, autoinflammation, and to attempt to limit neurologic disease): **baricitinib** and **ruxolitinib**; **tofacitinib** (JAK1/3) has shown benefit for inflammatory skin lesions in related interferonopathies. RNA-seq in ISG15-deficient patients nominated **baricitinib**

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NCIT: baricitinib (C123899), ruxolitinib (C79809), tofacitinib (C97662). CHEBI: baricitinib (CHEBI:145303), ruxolitinib (CHEBI:66919), tofacitinib (CHEBI:71200).

- **Caveat:** JAK inhibitors give limited/heterogeneous benefit on established **neurologic** manifestations, and **CNS calcification can progress despite early treatment**

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— likely due to poor CNS bioavailability.

- **Supportive/rehabilitative:** Wound care for skin ulcers; management of seizures; immunization counseling.
- **Advanced/experimental:** No approved gene/cell therapy; hematopoietic stem cell transplantation is not standard. Recombinant ISG15 or ISG15-blocking agents were proposed conceptually

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but are not clinical therapies.

- **Pharmacogenomics:** None specific established.

13. Prevention

- **Primary prevention:** In families with a known affected child, **avoid BCG vaccination** in at-risk newborns to prevent BCG disease; **genetic counseling** for consanguineous families.
- **Secondary prevention:** **Early molecular diagnosis** (WES/panel), **cascade/carrier testing**, and **prenatal/preimplantation genetic testing** where families opt for it.
- **Tertiary prevention:** Prompt anti-mycobacterial treatment; early JAK inhibition to limit inflammatory complications; surveillance neuroimaging and dermatologic care.
- **Counseling:** Autosomal recessive recurrence risk = **25%** for future offspring of carrier couples.
- **Immunization/public health:** The key public-health action is **withholding live BCG** in known-affected kindreds (BCG is routine in many endemic regions, which is why patients often present post-BCG).

14. Other Species / Natural Disease

- **Taxonomy:** Human disease (*Homo sapiens*, NCBI Taxon 9606). Orthologs studied in **mouse** (*Mus musculus*, Taxon 10090; gene *Isg15*, NCBI Gene 100038882).
- **Comparative biology — critical caveat:** In **mice**, ISG15/ISGylation is a **bona fide antiviral effector** of type I IFN, and mouse Usp18 does **not** require Isg15 for stabilization. In **humans**, ISGylation is **redundant for antiviral defense**, and the disease-defining role is **USP18 stabilization/IFN-I regulation**

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Thus the human phenotype (interferonopathy) is **not recapitulated** by *Isg15*-knockout mice, which instead show altered antiviral responses.

- **Natural disease in other species:** No well-characterized naturally occurring ISG15-deficiency disease reported in companion animals/wildlife (not in OMIA as a defined phenotype). No zoonotic component.

15. Model Organisms

- **Mouse (*Isg15*^{-/-}, *Ube1L/Uba7*^{-/-}):** Model the **antiviral ISGylation** role and hematopoietic effects

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but **do not reproduce the human USP18-dependent interferonopathy** — a major limitation. Resources: MGI, IMPC.

- **Human cellular models: Patient-derived fibroblasts, B-cell lines, PBMCs, and ISG15-knockout human epithelial/fibroblast lines** best recapitulate the human phenotype — showing hyperactive JAK/STAT, IFN-I-induced apoptosis, EMT, and impaired wound healing

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iPSC/organoid and spatial-transcriptomic ex vivo skin models have been used.

- **Applications:** Dissecting IFN-I regulation (USP18/ISG15/STAT2 axis), fibrosis mechanisms, and testing JAK-inhibitor efficacy.

Evidence Table (Key PMIDs)

PMID	Contribution	Evidence type
22859821	First description; MSMD via loss of secreted ISG15 → low IFN- γ ; ISGylation redundant for antiviral immunity	Human clinical + in vitro
29100055 / 32553163	Extracellular ISG15 receptor is LFA-1 (CD11a/CD18); drives IFN- γ release from NK/T cells	In vitro / mechanistic
25307056	Intracellular ISG15 stabilizes USP18; loss → IFN- α/β over-amplification, interferonopathy, calcification	Human clinical + mechanistic
32402279	Third (dermatologic) phenotype; cell-type-specific IFN-I signaling in skin	Human clinical + molecular
41743834	IFN-I-driven fibrosis: apoptosis, EMT, myofibroblast activation, impaired healing	Ex vivo + in vitro
37984483	Novel Y140X; divergent sibling phenotypes; JAK/STAT hyperactivation; baricitinib nominated	Human clinical + RNA-seq
39365299	Recurrent c.83T>A (L28Q) in unrelated Brazilian families; low IFN- γ ; transcriptomics	Human clinical + in silico
35258551	USP18 deficiency (I60N) — mechanistic/differential context	Human clinical
32092142	STAT2 R148Q phenocopy of USP18 deficiency — differential context	Human clinical
42643649	MSMD genetic heterogeneity (~22 genes, IFN- γ axis)	Review/case
38381212 / 41871482	JAK-inhibitor efficacy/limits in interferonopathies (CNS)	Clinical
23579383	ISG15 as secreted IFN- γ -inducing molecule; therapeutic concepts	Review
28165509	Crystal structure of USP18–ISG15; C-terminal Ubl domain specificity	Structural
20591702	Ube1L/ISGylation mouse hematopoietic model	Mouse model

Supported vs. Refuted Hypotheses

Supported: - ISG15 loss causes MSMD through impaired IFN- γ (secreted-ISG15 mechanism). ✓

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- ISG15 loss causes a type I interferonopathy via failed USP18 stabilization. ✓

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- Disease has three co-dominant phenotypes (infectious/neurologic/dermatologic). ✓

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- JAK1/2 inhibition is the rational therapy for the interferonopathy. ✓

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Refuted / negative findings: - Human ISG15 deficiency causes severe viral disease — **refuted** (no unusual viral susceptibility;

([P 22859821](#))

- Mouse *Isg15* knockouts model the human disease — **largely refuted** (species divergence; mice model antiviral role only). - Co-inherited MPO variants contribute to pathology — **refuted** in the reported kindred

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Limitations and Future Directions

- Ultra-rare disease: no population epidemiology, survival statistics, or QoL instrument data; evidence rests on case reports/small kindreds.
- Genotype–phenotype determinants of which phenotype manifests remain unexplained (modifiers, exposure).
- CNS-penetrant anti-IFN therapies are an unmet need; current JAK inhibitors inadequately protect the brain.
- Better human models (organoids, iPSC-derived neurons/microglia) are needed given mouse divergence.

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